COLLEGE OF INFORMATION SCIENCE AND ELECTRONIC ENGINEERING

(ISEE)

Information Engineering Curriculum

(Updated in January 2025)

1. Educational Objectives

Information Engineering encompasses telecommunications, networking, signal and information processing - all important and rapidly growing industries in today's information age. It provides students with an understanding of the entire stack of modern networked computers, from the design and architecture of the CPU in a smartphone, through the information theory and wireless protocols connecting it to the internet, to the operating systems and databases providing back-end support in the cloud.

2. Graduation Requirement

The curriculum leading to the degree of Bachelor of Information Engineering at ZJU requires **135** credits and is organized into required courses, technical electives, liberal education, Chinese language and society study for international students, and other electives.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce you to the opportunities and resources that our program can offer you as you work towards your career goals. They also provide you with the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENGR 100	Engineering Orientation	1

3.2 Foundational Mathematics and Science

These courses stress the fundamental mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	Credit
MATH 221	Calculus I	4
MATH 231	Calculus Ⅱ	3
MATH 241	Calculus III	4
MATH 257	Linear Algebra with Computational Applications	3
MATH 285	Intro Differential Equations	3
PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec& Mag	4
PHYS 213	Univ Physics: Thermal Physics	2
	Total required	**27**

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of information engineering.

Course Code	Course Name	Credit
ECE 110	Intro to Electronics	3
ECE 210	Analog Signal Processing	4
ECE 220	Computer Systems & Programming	4
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1

ECE 313	Probability with Eng App	3
ECE 329	Fields and Waves I	3
ECE 385	Digital Systems Laboratory	3
ECE 498SS	Signals and Systems	4
ECE 498PC	Principles of Communications	3
ECE 498PI	Project Internship	2
ECE 498FTP	Final Term Project	12
	Total required	**45**

4. Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop further in mathematics, science and a technical course of study in information engineering of considerable breadth and focus. Students must complete a minimum of **15** credits on technical electives.

Course Code	**Course Name**	**Credit**
CS 101	Intro to Computing: Engrg & Sci	3
CS 225	Data Structures	4
MATH213	Introduction to Discrete Mathematics	3
CHEM102	General Chemistry I	3
CHEM103	General Chemistry Lab I	1
CHEM104	General Chemistry II	3
CHEM105	General Chemistry Lab II	1
ECE 314	Probability in Engineering Lab	1
ECE 330	Power Circuits & Electromechanics	3
ECE 340	Semiconductor Electronics	3
ECE 342	Electronic Circuits	3

ECE 343	Electronic Circuits Laboratory	1
ECE 350	Fields and Waves II	3
ECE 364	Programing Methods for Machine Learning	3
ECE 365	Data Science and Engineering	3
ECE 374	Introduction to Algorithms & Models of Computation	4
ECE 485	MEMS Devices & Systems	3

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the liberation education requirements, students must take all courses for grade and complete courses based on the table below:

Category	**Credit**
1) Humanities & Arts (Two courses)	6
2) Social & Behavioral Sciences (Two courses)	6
3) Culture Studies	6
Western/Comparative Culture(s) (One course)	
Non-Western Culture(s) (One course)	
Total required	**18**

Appropriate choices in the Social and Behavioral Sciences and in the Humanities and Arts will ensure that these courses also fulfil the requirements in the areas of Western and Non-Western Cultures.

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define research specialties. Students are encouraged to take cross-discipline courses as free electives for **9（2021-2022）and 8(2023-)** credits.

5. Chinese Language and Society

These courses are compulsory for all international students. They introduce Chinese modern history, social development, government policies, etc., and help students improve their Chinese and understanding about China.

International students need to complete Chinese language study and Chinese society to fulfill the graduate requirement.

		2020-2022	**2023-**
Course Code	**Course Name**	**Credit**	**Credit**
CHIN 1001	Chinese Ⅰ	4	4
CHIN 1002	Chinese Ⅱ	5	4
CHIN 1003	Chinese Ⅲ	4	4
CHIN 1004	Chinese Ⅳ	4	4
CHIN1005	International Chinese Language Proficiency and Testing		2
CULT 2001	China Survey	3	3
	Total required	**20**	**21**